

Course 3182

3 Credits: 3

Program Core

Source-grounded

Climate responsive architecture

□ To identify the influence of tropical climates on architecture and apply it on the

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3182 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3182.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Architecture
Course code	3182
Course title	Climate responsive architecture
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:2, T:1, P:0) Periods per semester: 45

Item	Official information
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To identify the influence of tropical climates on architecture and apply it on the
- design of shelters for human comfort.

Course outcomes

CO	Official outcome	Study level
CO1	different types of tropical climates and factors 10 Understanding affecting site climate. Explain human thermal comfort factors and summarize how structural controls affect	See official syllabus
CO2	11 Understanding thermal design of buildings in order to attain human comfort Explain the effect of ventilation and air	See official syllabus
CO3	movement and other passive design techniques 11 Understanding in cooling buildings Summarize design features of shelters in tropical	See official syllabus
CO4	11 Understanding climates	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3 2 2 2
CO4	CO4 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	climates and factors affecting site climate.	3	As specified
Module 2	controls affect thermal design of buildings in order to attain human	3	As specified
Module 3	design techniques in cooling buildings	3	As specified
Module 4	Summarize design features of shelters in tropical climates	4	As specified

MODULE 1

climates and factors affecting site climate.

This module is presented from the official syllabus scope for Climate responsive architecture. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: climates and factors affecting site climate.
- Official duration: As specified

- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain principal climatic elements 2 Understanding Classify tropical climates and their

Explain principal climatic elements 2 Understanding Classify tropical climates and their is an official topic in Module 1 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain principal climatic elements 2 Understanding Classify tropical climates and their using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain principal climatic elements 2 understanding classify tropical climates and their should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain principal climatic elements 2 Understanding Classify tropical climates and their means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain principal climatic elements 2 Understanding Classify tropical climates and their definition/meaning, steps, application Technical terms English-

4 Understanding characteristics Summarize factors affecting site climate and

4 Understanding characteristics Summarize factors affecting site climate and is an official topic in Module 1 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding characteristics Summarize factors affecting site climate and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding characteristics summarize factors affecting site climate and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding characteristics Summarize factors affecting site climate and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-4 Understanding characteristics Summarize factors affecting site climate and **പരമ്പരാഗത കെട്ടിട നിർമ്മാണം** definition/meaning **പരമ്പരാഗത കെട്ടിട നിർമ്മാണം**. **പരമ്പരാഗത കെട്ടിട നിർമ്മാണം**, steps, application **പരമ്പരാഗത കെട്ടിട നിർമ്മാണം** **പരമ്പരാഗത കെട്ടിട നിർമ്മാണം**. Technical terms English-**പരമ്പരാഗത കെട്ടിട നിർമ്മാണം**.

4 Understanding urban climate Climate and weather - definitions; Principal climatic elements - Solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland climate - characteristics of all these climates. Site climate - Definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (Brief explanation only) Urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. Explain human thermal comfort factors and summarize how structural

4 Understanding urban climate Climate and weather - definitions; Principal climatic elements - Solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland climate - characteristics of all these climates. Site climate - Definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (Brief explanation only) Urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. Explain human thermal comfort factors and summarize how structural is an official topic in Module 1 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding urban climate Climate and weather - definitions; Principal climatic elements - Solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland climate - characteristics of all these climates. Site climate - Definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (Brief explanation only) Urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. Explain human thermal comfort factors and summarize how structural using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding urban climate climate and weather - definitions; principal climatic elements - solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - trade winds - thermal forces, westerlies and polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - tilt of the earth's axis - earth's thermal balance - sky conditions, vegetation classification of tropical climates - tropical climates - warm humid climate - warm humid island climate - hot dry desert climate - hot dry maritime desert climate - composite or monsoon climate - tropical upland climate - characteristics of all these climates. site climate - definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (brief explanation only) urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. explain human thermal comfort factors and summarize how structural should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding urban climate Climate and weather - definitions; Principal climatic elements - Solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind

Aspect	What to prepare
	pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland climate - characteristics of all these climates. Site climate - Definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (Brief explanation only) Urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. Explain human thermal comfort factors and summarize how structural means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding urban climate Climate and weather - definitions; Principal climatic elements - Solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland climate - characteristics of all these climates. Site climate - Definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (Brief explanation only) Urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. Explain human thermal comfort factors and summarize how structural definition/meaning. , steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

climates and factors affecting site climate.

M1.01	Explain principal climatic elements	2
Understanding	Classify tropical climates and their	
M1.02		4
Understanding	characteristics	
	Summarize factors affecting site climate and	
M1.03		4
Understanding	urban climate	

Contents:

Climate and weather - definitions; Principal climatic elements - Solar radiation-quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation
Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland climate - characteristics of all these

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

controls affect thermal design of buildings in order to attain human

This module is presented from the official syllabus scope for Climate responsive architecture. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: controls affect thermal design of buildings in order to attain human
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

2 Understanding regulatory mechanisms of human body

2 Understanding regulatory mechanisms of human body is an official topic in Module 2 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding regulatory mechanisms of human body using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding regulatory mechanisms of human body should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding regulatory mechanisms of human body means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding regulatory mechanisms of human body ഓരോ ഓരോ definition/meaning ഓരോ ഓരോ. ഓരോ ഓരോ, steps, application ഓരോ ഓരോ. Technical terms English- ഓരോ ഓരോ.

Explain heat exchange process of buildings 2 Understanding Identify the effect of structural controls on

Explain heat exchange process of buildings 2 Understanding Identify the effect of structural controls on is an official topic in Module 2 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain heat exchange process of buildings 2 Understanding Identify the effect of structural controls on using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain heat exchange process of buildings 2 understanding identify the effect of structural controls on should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain heat exchange process of buildings 2 Understanding Identify the effect of structural controls on means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നിരിക്കട്ടെ Explain heat exchange process of buildings 2 Understanding Identify the effect of structural controls on ഏകദേശം ഏകദേശം definition/meaning ഏകദേശം ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-എന്നിരിക്കട്ടെ ഏകദേശം ഏകദേശം.

7 Understanding thermal design of buildings Body's heat production - heat loss - Regulatory mechanisms - Subjective variables. Heat exchange processes of buildings - thermal balance equation, thermal design. Thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other special glasses, external shading devices - sun's positions-angle of incidence - shadow angles - shading devices - design of shading devices. Explain the effect of ventilation and air movement and other passive

7 Understanding thermal design of buildings Body's heat production - heat loss - Regulatory mechanisms - Subjective variables. Heat exchange processes of buildings - thermal balance equation, thermal design. Thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other special glasses, external shading devices - sun's positions-angle of incidence - shadow angles - shading devices - design of shading devices. Explain the effect of ventilation and air movement and other passive is an official topic in Module 2 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Understanding thermal design of buildings Body's heat production - heat loss - Regulatory mechanisms - Subjective variables. Heat exchange processes of buildings - thermal balance equation, thermal design. Thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other special glasses, external shading devices - sun's positions-angle of incidence - shadow angles - shading devices - design of shading devices. Explain the effect of ventilation and air movement and other passive using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 understanding thermal design of buildings body's heat production - heat loss - regulatory mechanisms - subjective variables. heat exchange processes of buildings - thermal balance equation, thermal design. thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other special glasses, external shading devices - sun's positions-angle of incidence - shadow angles - shading devices - design of shading devices. explain the effect of ventilation and air movement and other passive should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Understanding thermal design of buildings Body's heat production - heat loss - Regulatory mechanisms - Subjective variables. Heat exchange processes of buildings - thermal balance equation, thermal design. Thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other special glasses, external shading devices - sun's positions-angle of incidence - shadow angles - shading devices - design of shading devices. Explain the effect of ventilation and air movement and other passive means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 7 Understanding thermal design of buildings Body's heat production - heat loss - Regulatory mechanisms - Subjective variables. Heat exchange processes of buildings - thermal balance equation, thermal design. Thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other

Official syllabus detail and terminology

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

design techniques in cooling buildings

This module is presented from the official syllabus scope for Climate responsive architecture. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: design techniques in cooling buildings
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain the effect of ventilation and air 6 Understanding movement on thermal design of buildings Identify passive cooling techniques in

Explain the effect of ventilation and air 6 Understanding movement on thermal design of buildings Identify passive cooling techniques in is an official topic in Module 3 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the effect of ventilation and air 6 Understanding movement on thermal design of buildings Identify passive cooling techniques in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the effect of ventilation and air 6 understanding movement on thermal design of buildings identify passive cooling techniques in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the effect of ventilation and air 6 Understanding movement on thermal design of buildings Identify passive cooling techniques in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain the effect of ventilation and air 6
Understanding movement on thermal design of buildings Identify passive cooling
techniques in . definition/meaning .
 , steps, application . Technical terms
English- .

3 Understanding buildings Identify passive cooling methods adopted

3 Understanding buildings Identify passive cooling methods adopted is an official topic in Module 3 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding buildings Identify passive cooling methods adopted using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding buildings identify passive cooling methods adopted should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding buildings Identify passive cooling methods adopted means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 3 Understanding buildings Identify passive cooling methods adopted ഓരോന്നിന്റെയും definition/meaning എഴുതുക. ഓരോന്നിന്റെയും steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

in a traditional building and make a report 2 Applying after conducting literature study Ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings- humidity control-working of wind - scoop. Passive design - Passive cooling - orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - Envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel.

in a traditional building and make a report 2 Applying after conducting literature study Ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings- humidity control-working of wind - scoop. Passive design - Passive cooling - orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - Envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel. is an official topic in Module 3 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify in a traditional building and make a report 2 Applying after conducting literature study Ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings- humidity control-working of wind -

scoop. Passive design - Passive cooling – orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - Envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, in a traditional building and make a report 2 applying after conducting literature study ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings- humidity control-working of wind - scoop. passive design - passive cooling – orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What in a traditional building and make a report 2 Applying after conducting literature study Ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings- humidity control-working of wind - scoop. Passive design - Passive cooling – orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - Envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- in a traditional building and make a report 2 Applying after conducting literature study Ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings- humidity control- working of wind - scoop. Passive design - Passive cooling - orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - Envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

design techniques in cooling buildings

M3.01 Explain the effect of ventilation and air

6

Understanding

movement on thermal design of buildings

Identify passive cooling techniques in

M3.02

3

Understanding

buildings

Identify passive cooling methods adopted

M3.03

2

Applying

in a traditional building and make a report

after conducting literature study

Contents:

Ventilation and air movement - functions of ventilation - supply of fresh air, convective

cooling - provision for ventilation - stack effect - physiological cooling-provision for air

movement - wind effects - air flow through buildings - orientation, cross ventilation, size

of openings, controls of openings - air movement and rain - air flow around buildings-

humidity control-working of wind - scoop.

Passive design - Passive cooling – orientation - ventilation - convective cooling - air

movement - physiological and evaporative cooling, - shading - shading devices,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Summarize design features of shelters in tropical climates

This module is presented from the official syllabus scope for Climate responsive architecture. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize design features of shelters in tropical climates
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding hot dry climate Summarize design features of shelters suitable

3 Understanding hot dry climate Summarize design features of shelters suitable is an official topic in Module 4 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding hot dry climate Summarize design features of shelters suitable using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding hot dry climate summarize design features of shelters suitable should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding hot dry climate Summarize design features of shelters suitable means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding hot dry climate Summarize design features of shelters suitable for hot dry climate. definition/meaning of shelter. steps, application of shelter. Technical terms English-Malayalam.

for warm humid climate Summarize design features of shelters suitable

for warm humid climate Summarize design features of shelters suitable is an official topic in Module 4 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify for warm humid climate Summarize design features of shelters suitable using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, for warm humid climate summarize design features of shelters suitable should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What for warm humid climate Summarize design features of shelters suitable means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- for warm humid climate Summarize design features of shelters suitable for warm humid climate. definition/meaning of shelter. steps, application of shelter. Technical terms English-Malayalam.

3 Understanding for composite climate Summarize design features of shelters suitable

3 Understanding for composite climate Summarize design features of shelters suitable is an official topic in Module 4 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding for composite climate Summarize design features of shelters suitable using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding for composite climate summarize design features of shelters suitable should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding for composite climate Summarize design features of shelters suitable means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding for composite climate
Summarize design features of shelters suitable for composite climate
definition/meaning. Explain the concept, steps, application
of the concept. Technical terms English-Malayalam.

2 Understanding for tropical upland climate Shelter for hot - dry climate- nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow - traditional shelter. Shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter. Shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings - ventilation and condensation - traditional shelter. Shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls | surface treatment - openings - traditional shelter. Text / Reference: T/R Book Title/Author Koenigsberger, Ingersoll, Mayhew and Szokolay - MANUAL OF TROPICAL T1 HOUSING AND BUILDING R2 Givoni - Man, climate and architecture Online resources: Sl. No Website Link

https://www.researchgate.net/publication/312432251_An_overview_of_passive_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions 1

<http://www.yourhome.gov.au/passive-design/passive-cooling> 2

<https://www.greenroofplan.com/intensive-vs-extensive-green-roofs> 3

https://en.wikipedia.org/wiki/Passive_cooling 4

https://en.wikipedia.org/wiki/Solar_chimney 5

2 Understanding for tropical upland climate Shelter for hot - dry climate-nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow - traditional shelter. Shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter. Shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings - ventilation and condensation - traditional shelter. Shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls | surface treatment - openings - traditional shelter. Text / Reference: T/R Book Title/Author Koenigsberger, Ingersoll, Mayhew and Szokolay - MANUAL OF TROPICAL T1 HOUSING AND BUILDING R2 Givoni - Man, climate and architecture Online resources: Sl. No Website Link

https://www.researchgate.net/publication/312432251_An_overview_of_passive_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions 1

<http://www.yourhome.gov.au/passive-design/passive-cooling> 2

<https://www.greenroofplan.com/intensive-vs-extensive-green-roofs> 3

<https://www.greenroofplan.com/intensive-vs-extensive-green-roofs> 4

https://en.wikipedia.org/wiki/Passive_cooling 5

https://en.wikipedia.org/wiki/Solar_chimney is an official topic in Module 4 of Climate responsive architecture. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding for tropical upland climate Shelter for hot - dry climate-nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow - traditional shelter. Shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter. Shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings - ventilation and condensation - traditional shelter. Shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls | surface treatment - openings - traditional shelter. Text / Reference: T/R Book Title/Author Koenigsberger, Ingersoll, Mayhew and Szokolay - MANUAL OF TROPICAL T1 HOUSING AND BUILDING R2 Givoni - Man, climate and architecture Online resources: SI. No Website Link
https://www.researchgate.net/publication/312432251_An_overview_of_passive_1_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions 2
<http://www.yourhome.gov.au/passive-design/passive-cooling> 3
<https://www.greenroofplan.com/intensive-vs-extensive-green-roofs> 4
https://en.wikipedia.org/wiki/Passive_cooling 5
https://en.wikipedia.org/wiki/Solar_chimney using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding for tropical upland climate shelter for hot - dry climate-nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow -

traditional shelter. shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter. shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings - ventilation and condensation - traditional shelter. shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls | surface treatment - openings - traditional shelter. text / reference: t/r book title/author koenigsberger, ingersoll, mayhew and szokolay - manual of tropical t1 housing and building r2 givoni - man, climate and architecture online resources: sl. no website link
https://www.researchgate.net/publication/312432251_an_overview_of_passive_1_cooling_techniques_in_buildings_design_concepts_and_architectural_interventions 2
<http://www.yourhome.gov.au/passive-design/passive-cooling> 3
<https://www.greenroofplan.com/intensive-vs-extensive-green-roofs> 4
https://en.wikipedia.org/wiki/passive_cooling 5
https://en.wikipedia.org/wiki/solar_chimney should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding for tropical upland climate Shelter for hot - dry climate-nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow - traditional shelter. Shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter. Shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings - ventilation and condensation - traditional shelter. Shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls surface treatment - openings - traditional shelter. Text / Reference: T/R Book Title/Author Koenigsberger, Ingersoll, Mayhew and Szokolay - MANUAL OF TROPICAL T1 HOUSING AND BUILDING R2 Givoni - Man, climate and architecture Online resources: Sl. No Website Link https://www.researchgate.net/publication/312432251_An_overview_of_passive_1_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions 2 http://www.yourhome.gov.au/passive-design/passive-cooling 3 https://www.greenroofplan.com/intensive-vs-extensive-green-roofs 4 https://en.wikipedia.org/wiki/Passive_cooling 5 https://en.wikipedia.org/wiki/Solar_chimney means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- 2 Understanding for tropical upland climate

Shelter for hot - dry climate-nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow - traditional shelter. Shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter. Shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings -ventilation and condensation - traditional shelter. Shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls | surface treatment - openings - traditional shelter.

Text / Reference: T/R Book Title/Author Koenigsberger, Ingersoll, Mayhew and Szokolay - MANUAL OF TROPICAL T1 HOUSING AND BUILDING R2 Givoni - Man, climate and architecture Online resources: Sl. No Website Link

https://www.researchgate.net/publication/312432251_An_overview_of_passive_1_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions 2 <http://www.yourhome.gov.au/passive-design/passive-cooling> 3

<https://www.greenroofplan.com/intensive-vs-extensive-green-roofs> 4

https://en.wikipedia.org/wiki/Passive_cooling 5

https://en.wikipedia.org/wiki/Solar_chimney 6

definition/meaning . , steps, application . Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Summarize design features of shelters in tropical climates		
Outline design features of shelters suitable for		
M4.01		3
Understanding	hot dry climate	
	Summarize design features of shelters suitable	3
Understanding		
M4.02	for warm humid climate	
	Summarize design features of shelters suitable	
M4.03		3
Understanding	for composite climate	
	Summarize design features of shelters suitable	
M4.04		2
Understanding	for tropical upland climate	
	Series Test -II	1
Contents:		
Shelter for hot - dry climate-nature of the climate - physiological objectives - form and		
planning - external spaces - roofs, walls and openings - roof and wall surfaces		
- ventilation		
and air flow - traditional shelter.		
Shelter for warm humid climates - nature of the climate - physiological		
objectives - form		
and planning external spaces - roofs and walls - air flow and openings -		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain principal climatic elements 2 Understanding Classify tropical climates and their. (3 marks)

Answer: Begin with the standard definition or identity of *Explain principal climatic elements 2 Understanding Classify tropical climates and their*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding characteristics Summarize factors affecting site climate and. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding characteristics Summarize factors affecting site climate and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding urban climate Climate and weather - definitions; Principal climatic elements - Solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland climate - characteristics of all these climates. Site climate - Definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (Brief explanation only) Urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. Explain human thermal comfort factors and summarize how structural. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding urban climate Climate and weather - definitions; Principal climatic elements - Solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland*

climate - characteristics of all these climates. Site climate - Definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (Brief explanation only) Urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. Explain human thermal comfort factors and summarize how structural. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Understanding regulatory mechanisms of human body. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding regulatory mechanisms of human body*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Explain heat exchange process of buildings 2 Understanding Identify the effect of structural controls on. (3 marks)

Answer: Begin with the standard definition or identity of *Explain heat exchange process of buildings 2 Understanding Identify the effect of structural controls on*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 7 Understanding thermal design of buildings Body's heat production - heat loss - Regulatory mechanisms - Subjective variables. Heat exchange processes of buildings - thermal balance equation, thermal design. Thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other special glasses, external shading devices - sun's positions-angle of incidence - shadow angles - shading devices - design of shading devices. Explain the effect of ventilation and air movement and other passive. (3 marks)

Answer: Begin with the standard definition or identity of *7 Understanding thermal design of buildings Body's heat production - heat loss - Regulatory mechanisms - Subjective variables. Heat exchange processes of buildings - thermal balance equation, thermal design. Thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other special glasses, external shading devices - sun's positions-angle of incidence - shadow angles - shading devices - design of shading devices. Explain the effect of ventilation and air movement and other passive.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Explain the effect of ventilation and air 6 Understanding movement on thermal design of buildings Identify passive cooling techniques in. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the effect of ventilation and air 6 Understanding movement on thermal design of buildings Identify passive cooling techniques in.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding buildings Identify passive cooling methods adopted. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding buildings Identify passive cooling methods adopted*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain in a traditional building and make a report 2 Applying after conducting literature study Ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings- humidity control-working of wind - scoop. Passive design - Passive cooling - orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - Envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel.. (3 marks)

Answer: Begin with the standard definition or identity of *in a traditional building and make a report 2 Applying after conducting literature study Ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings- humidity control-working of wind - scoop. Passive design - Passive cooling - orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - Envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel..* State the governing principle, list the key components or stages, and close with one

relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain 3 Understanding hot dry climate Summarize design features of shelters suitable. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding hot dry climate Summarize design features of shelters suitable*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain for warm humid climate Summarize design features of shelters suitable. (3 marks)

Answer: Begin with the standard definition or identity of *for warm humid climate Summarize design features of shelters suitable*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding for composite climate Summarize design features of shelters suitable. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding for composite climate Summarize design features of shelters suitable*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding for tropical upland climate Shelter for hot - dry climate-nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow - traditional shelter. Shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter. Shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings -ventilation and condensation - traditional shelter. Shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls | surface treatment - openings - traditional shelter. Text / Reference: T/R Book Title/Author Koenigsberger, Ingersoll, Mayhew and Szokolay - MANUAL OF TROPICAL T1 HOUSING AND BUILDING R2 Givoni - Man, climate and architecture Online resources: Sl. No Website Link
https://www.researchgate.net/publication/312432251_An_overview_of_passive_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions
1
2 <http://www.yourhome.gov.au/passive-design/passive-cooling>
3 <https://www.greenroofplan.com/intensive-vs-extensive-green-roofs>
4 https://en.wikipedia.org/wiki/Passive_cooling
5 https://en.wikipedia.org/wiki/Solar_chimney. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding for tropical upland climate Shelter for hot - dry climate-nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow - traditional shelter. Shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter. Shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings - ventilation and condensation - traditional shelter. Shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls | surface treatment - openings - traditional shelter. Text / Reference: T/R Book Title/Author Koenigsberger, Ingersoll, Mayhew and Szokolay - MANUAL OF TROPICAL T1 HOUSING AND BUILDING R2 Givoni - Man, climate and architecture Online resources: Sl. No Website Link*
https://www.researchgate.net/publication/312432251_An_overview_of_passive_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions
1

cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions 2

<http://www.yourhome.gov.au/passive-design/passive-cooling> 3

<https://www.greenroofplan.com/intensive-vs-extensive-green-roofs> 4

https://en.wikipedia.org/wiki/Passive_cooling 5

https://en.wikipedia.org/wiki/Solar_chimney. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain principal climatic elements 2 Understanding Classify tropical climates and their.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding regulatory mechanisms of human body.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain the effect of ventilation and air 6 Understanding movement on thermal design of buildings Identify passive cooling techniques in.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 Understanding hot dry climate Summarize design features of shelters suitable.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.

- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link
- https://www.researchgate.net/publication/312432251_An_overview_of_passive_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventi
- <http://www.yourhome.gov.au/passive-design/passive-cooling>
- <https://www.greenroofplan.com/intensive-vs-extensive-green-roofs>
- https://en.wikipedia.org/wiki/Passive_cooling
- https://en.wikipedia.org/wiki/Solar_chimney

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTR syllabus PDF for Course 3182. Verify institutional updates against the current official curriculum.